Problem

In the circuit of Fig. 3.47, the voltage $v2$ is:

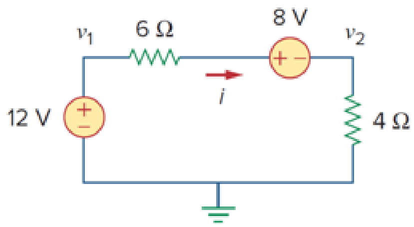
(a) −8 V

(b) −1.6 V

(c) 1.6 V

(d) 8 V

Figure 3.47

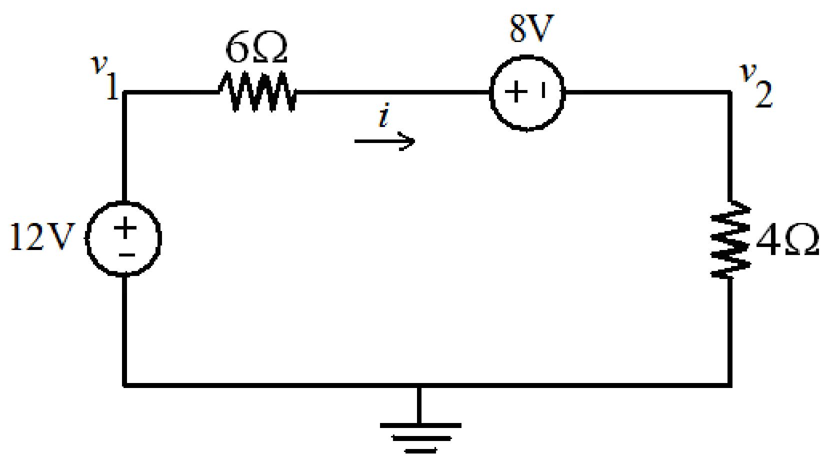


Step-by-step solution

Step 1 of 3

Answer: (c)

Given circuit diagram



Step 2 of 3

From the circuit by using KVL we can write

$$-12+6i+8+4i=0$$

$$10i=4$$

$$i=\frac{4}{10}$$

$$i=0.4\text{A}$$

Step 3 of 3

Voltage $v_2 = 4i$

$$v_2 = 4(0.4)$$

$$v_2 = 1.6\text{V}$$